# MULTIPATH LOAD BALANCING USING RYU CONTROLLER

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*in*

***Electronics and Communication Engineering***

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***Computer Science and Engineering***

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## CERTIFICATE

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## DECLARATION

I Hridya Sreekumar hereby declare that the project report **Multipath load balancing using RYU controller**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of Smt. Soumya M. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources.

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## ABSTRACT

The exponential growth of network traffic, driven by technologies like 5G, IoT, and cloud computing, has placed immense pressure on traditional network architectures. Ensuring high Quality of Service (QoS) parameters such as low latency, minimal jitter, and negligible packet loss is a significant challenge. Load balancing is a critical technique to distribute traffic across multiple paths, preventing congestion, optimizing resource utilization, and enhancing overall network performance. However, traditional networking paradigms, with their distributed control planes, often struggle to implement dynamic and efficient load-balancing policies.

Software-Defined Networking (SDN) emerges as a transformative solution by decoupling the network's control logic (control plane) from the underlying forwarding devices (data plane). This centralization of intelligence in a software-based controller offers a global view of the network, enabling the implementation of sophisticated, dynamic traffic management strategies.

This project report details the design, implementation, and evaluation of an intelligent, **Bandwidth-aware Multipath Load Balancer** using the **Ryu SDN controller**. The core of the system is a dynamic algorithm that discovers all available paths between a source and destination using a **Depth-First Search (DFS)** algorithm. Unlike traditional methods like Shortest Path or Equal-Cost Multi-Path (ECMP) routing, our solution calculates the most optimal path based on real-time available bandwidth, effectively routing traffic away from congested links.

The proposed system was implemented and tested on a custom network topology emulated using **Mininet**. The Ryu controller application was developed in Python, leveraging the OpenFlow protocol for communication with the data plane switches. Performance was rigorously evaluated against standard algorithms by measuring key QoS metrics: **end-to-end delay, jitter, and packet loss rate**.

The experimental results conclusively demonstrate that the proposed bandwidth-aware algorithm significantly outperforms its counterparts. It achieved a notable reduction in delay (up to 48% compared to Shortest Path), more stable jitter (40-43% lower than ECMP), and a substantially lower packet loss rate (approximately 58% lower than Shortest Path). This project successfully validates the efficacy of the SDN paradigm and the Ryu framework in developing advanced, responsive, and efficient traffic engineering solutions for modern high-demand networks.

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# Chapter 1

# Introduction

## 1.1 Background and Motivation

The digital landscape is undergoing a radical transformation with the proliferation of data-intensive applications, Internet of Things (IoT) devices, and high-definition streaming services. This has led to an unprecedented surge in network traffic, pushing traditional network infrastructures to their limits. Conventional networks, characterized by a distributed architecture where control and data planes are embedded within each networking device (e.g., routers, switches), are inherently rigid. This rigidity makes it exceptionally challenging to implement network-wide policies, adapt to changing traffic patterns in real-time, and optimize resource utilization dynamically.

Software-Defined Networking (SDN) has emerged as a paradigm shift, addressing these limitations by introducing a centralized, programmable approach to network management. By decoupling the control plane (the "brain" that makes forwarding decisions) from the data plane (the "muscle" that forwards packets), SDN provides a holistic view of the network. This global perspective enables network administrators to manage the entire network from a single point and deploy innovative applications for traffic engineering, security, and automation. Load balancing, a critical component of traffic engineering, stands to benefit immensely from this centralized intelligence.

## 1.2 Problem Statement

In traditional IP networks, load balancing is often handled by static protocols or simple algorithms. For instance, the Equal-Cost Multi-Path (ECMP) protocol distributes flows across multiple paths that have an identical cost (e.g., hop count). However, ECMP is oblivious to the real-time state of the network. It does not consider dynamic parameters such as current link utilization, available bandwidth, or latency. Consequently, it can easily lead to situations where one path becomes congested while another remains underutilized, resulting in increased packet loss, higher latency, and degraded Quality of Service (QoS). There is a clear need for a more intelligent, adaptive load-balancing mechanism that can react to real-time network conditions.

### 1.3 Objectives of the Project

The primary objectives of this project are:

1. To analyze the limitations of existing load-balancing techniques in traditional and SDN environments.
2. To design a dynamic multipath load-balancing algorithm for SDN that utilizes real-time bandwidth information for path selection.
3. To implement the proposed algorithm as a custom application on the Ryu SDN controller.
4. To emulate a network topology using Mininet to create a realistic testing environment.
5. To evaluate the performance of the implemented solution by comparing it against standard algorithms (Shortest Path and ECMP) based on critical QoS metrics: Delay, Jitter, and Packet Loss.

### 1.4 Scope of the Project

This project is scoped to the emulation environment provided by Mininet. The implementation focuses on a specific load-balancing algorithm based on available bandwidth. The network topology used for testing is a custom-built topology designed to demonstrate multipath capabilities. The evaluation is limited to the metrics of delay, jitter, and packet loss for TCP traffic.

### 1.5 Report Organization

This report is structured into six chapters. Following this introduction, Chapter 2 provides a literature survey and background on relevant technologies. Chapter 3 details the system analysis and design of the proposed solution. Chapter 4 elaborates on the implementation process. Chapter 5 presents and discusses the results of the performance evaluation. Finally, Chapter 6 concludes the report and suggests potential future enhancements.

# Chapter 2

# Literature survey and technology background

### 2.1 Evolution of Software-Defined Networking (SDN)

The concept of SDN originated from the need for more manageable, adaptable, and cost-effective networks. Key research projects like Ethane at Stanford University laid the groundwork by proposing the separation of the control and data planes. This evolution marks a move from distributed, hardware-centric networks to centralized, software-driven network control.

### 2.2 SDN Architecture

The SDN architecture is typically conceptualized in three distinct layers:

- **Data Plane:** Consists of network elements (switches, routers) that are responsible for forwarding traffic based on rules installed by the control plane. These are simple forwarding devices.
- **Control Plane:** A logically centralized software entity, the SDN controller, manages the network. It has a global view of the topology and is responsible for making all forwarding decisions.
- **Application Plane:** Network applications and services (e.g., load balancers, firewalls) run on top of the controller, using its Northbound API to define network behavior.

### 2.3 The OpenFlow Protocol

OpenFlow is the first and most widely deployed standard communication protocol between the control and data planes in an SDN architecture. It allows the controller to remotely configure the flow tables of OpenFlow-enabled switches. A flow table contains match fields (e.g., source IP, destination IP) and corresponding actions (e.g., forward to port, drop). When a switch receives a packet that does not match any existing flow entry, it encapsulates the packet and sends it to the controller as a Packet_In message. The controller then decides how to handle the packet and can install new flow entries to handle similar subsequent packets at line rate.

### 2.4 Load Balancing: Concepts and Challenges

Load balancing is the process of distributing network traffic across multiple paths to ensure that no single link, switch, or server becomes a bottleneck. The key challenges include:

- **Avoiding Congestion:** Distributing traffic to prevent any path from being overloaded.
- **Maximizing Throughput:** Utilizing all available network resources to their fullest.
- **Minimizing Latency:** Ensuring packets take the fastest available route.
- **Maintaining Stability:** Avoiding frequent path changes that can cause flapping and instability.

### 2.5 Multipath Routing Strategies

- **Shortest Path Routing (Dijkstra):** This algorithm finds a single path with the lowest cost (e.g., number of hops). Its main drawback is its static nature; it does not adapt to current network load, potentially overloading the shortest path.
- **Equal-Cost Multi-Path (ECMP):** ECMP improves upon shortest-path routing by distributing flows across multiple paths that have the same cost. However, it is "flow-based" and not "packet-based," and more importantly, it is unaware of the dynamic capacity of each path.
- **Weighted and Dynamic Multipath Routing:** Advanced algorithms assign weights to paths based on dynamic metrics like available bandwidth, delay, or loss. Traffic is distributed in proportion to these weights, leading to more efficient utilization. The algorithm implemented in this project falls into this category.

### 2.6 Review of Related Work

Substantial research has been conducted in the domain of SDN-based load balancing. The work most directly relevant to this project is by Chahlaoui et al. (2022), who proposed a bandwidth-based multipath load-balancing mechanism. Their simulations demonstrated superior performance in delay, jitter, and packet loss compared to ECMP. Other notable works include:

- **MPF-MLBS (2020):** A strategy based on Multiple Performance Factors (delay, bandwidth, link rate).
- **W-ECMP (2018):** A weighted version of ECMP programmed into the data plane.
- **DAMR (2018):** A Dynamic and Adaptive Multi-path Routing algorithm sensitive to packet loss and jitter.

This project draws inspiration from these concepts and provides a practical implementation of a bandwidth-aware algorithm using the Ryu controller.

# Chapter 3

# System analysis and design

### 3.1 Problem Analysis

The core problem identified is the inefficiency of static load-balancing algorithms in dynamically changing network environments. A solution must be:

- **Dynamic:** Capable of adapting to real-time network state.
- **Global:** Able to make decisions based on a network-wide view.
- **Efficient:** Should not introduce significant overhead. SDN's centralized controller is the ideal platform to host such a solution.

### 3.2 Proposed System Architecture

The system follows a classic SDN three-tier architecture:

1. **Application Layer:** Hosts the custom BandwidthLoadBalancer Ryu application.
2. **Control Layer:** The Ryu controller framework, which provides the Northbound API for the application and the Southbound OpenFlow interface for the data plane.
3. **Infrastructure Layer:** The Mininet-emulated network of OpenFlow switches and hosts

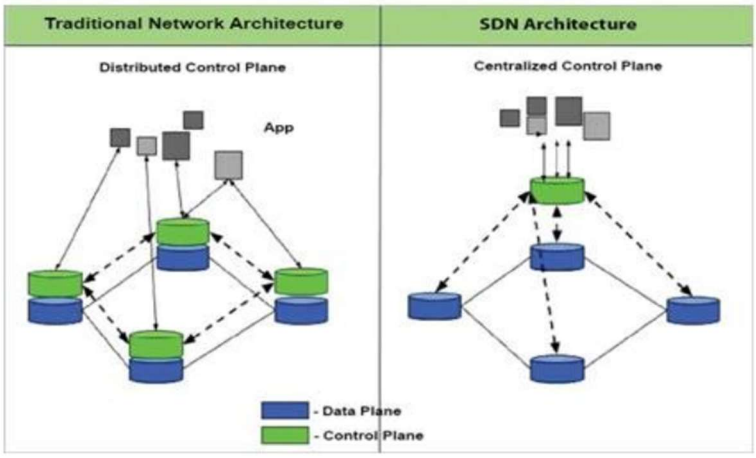


Figure 3.1: SDN-based Load Balancing System Architecture

### 3.3 Design Methodology

An object-oriented design methodology was employed. Key components were modeled as Python classes and modules within the Ryu application, ensuring modularity and maintainability.

### 3.4 The Bandwidth-Based Load Balancing Algorithm

The algorithm operates in three distinct phases:

#### 3.4.1 Path Discovery Phase

- **Input:** A graph G representing the network topology, a source node s, and a destination node d.
- **Process:** A non-recursive Depth-First Search (DFS) algorithm is used to traverse the graph and find all possible acyclic paths from s to d.
- **Output:** A list P containing all discovered paths. DFS is preferred for its memory efficiency in exploring deep branches, which is suitable for network path discovery.

#### 3.4.2 Bandwidth Calculation Phase

- For each path p in the list P:
    1. For each link l in path p, the controller polls the switch ports to get the number of bytes transmitted (byte_count).
    2. The used bandwidth for link l over a time interval Δt is calculated as: [ used_bw_l = (byte_count_t2 - byte_count_t1) / Δt ]
    3. The available bandwidth for link l is: [ avail_bw_l = total_capacity_l - used_bw_l ]
    4. The available bandwidth for the entire path p is determined by its bottleneck link:
       [ avail_bw_p = min(avail_bw_l for all l in p) ]

#### 3.4.3 Path Selection and Flow Installation Phase

- The path with the **maximum** avail_bw_p is selected as the optimal path for the new flow.
- The controller computes the sequence of OpenFlow commands to install flow entries on every switch along the selected path. These entries match the flow's characteristics (e.g., IP addresses) and specify the output port for forwarding.
- The initial packet that triggered the Packet_In event is then forwarded out along this path.

### 3.5 System Workflow

The sequence of operations is as follows:

1. Host A sends the first packet of a new flow to Host B.
2. Switch 1 receives the packet, finds no matching flow entry, and sends a Packet_In to

the Ryu controller.

3. The controller's load balancer application is invoked.
4. The application runs the three-phase algorithm (Discover -> Calculate -> Select).
5. The controller installs flow entries on all switches along the best path using Flow_Mod messages.
6. The controller sends the buffered packet out of the appropriate port on Switch 1 using a Packet_Out message.
7. All subsequent packets of this flow are switched at line speed along the installed path without controller intervention.

# Chapter 4

# Implementation details

### 4.1 Tools and Technologies Used

- **Mininet 2.3.0:** A network emulator that creates a virtual network of hosts, switches, and links on a single machine. It was used to create and test the custom topology.
- **Ryu SDN Controller 4.34:** A component-based SDN framework written in Python. It was chosen for its well-documented API, active community, and ease of use for developing custom applications.
- **OpenFlow 1.3:** The version of the OpenFlow protocol used for communication between Ryu and the Mininet switches.
- **Python 3.8:** The primary programming language for writing the Ryu application.
- **Iperf:** A tool for generating TCP traffic and measuring bandwidth performance.

### 4.2 Development Environment Setup

The entire project was developed on a virtual machine running Ubuntu 20.04 LTS. The environment was set up by installing Mininet from source and Ryu via the Python package manager (pip).

### 4.3 Topology Design and Emulation

A custom linear topology with multiple parallel paths was created using Mininet's Python API. This topology was designed to have at least two distinct paths between the source and destination hosts to facilitate multipath routing. All links were configured with a bandwidth of 10 Mbps to create a constrained environment where load-balancing decisions would have a visible impact.

### 4.4 Ryu Application Module Development

The custom Ryu application, load_balancer.py, was structured into several logical modules:

- **Topology Discovery Module:** This module uses Link Layer Discovery Protocol (LLDP) packets to automatically discover and maintain a graph of the network topology. It listens for Packet_In events containing LLDP packets to identify links

between switches.

- **Statistics Collection Module:** This module periodically (e.g., every 5 seconds) sends PortStats requests to all switches in the network. The received statistics (e.g., rx_bytes, tx_bytes) are stored and used to calculate the used bandwidth on each link.
- **Core Load Balancer Module:** This is the main module that handles Packet_In events for data packets. It integrates the topology information from the first module and the bandwidth statistics from the second module to execute the three-phase load-balancing algorithm. It is responsible for making the path selection decision and issuing the Flow_Mod and Packet_Out commands.

### 4.5 Testing and Validation Strategy

Testing was an iterative process. Initial tests verified that the topology was discovered correctly. Subsequent tests involved generating low-volume ping traffic to ensure basic connectivity and path installation. Finally, high-volume TCP traffic was generated using Iperf to stress the network and collect meaningful performance data for the comparative analysis.

# Chapter 5

# Results and performance evaluation

### 5.1 Experimental Setup and Parameters

- **Topology:** A custom Mininet topology with 2 parallel paths.
- **Link Capacity:** 10 Mbps for all links.
- **Traffic:** TCP traffic generated using Iperf, with data rates varying from 10 Mbps to 60 Mbps to simulate low to high load conditions.
- **Compared Algorithms:**
    1. **Proposed Algorithm:** Bandwidth-based Multipath Load Balancer.
    2. **ECMP:** Standard Equal-Cost Multi-Path routing.
    3. **Shortest Path:** Dijkstra's single shortest-path algorithm.
- **Metrics Measured:** Average End-to-End Delay (ms), Jitter (ms), Packet Loss Rate.

### 5.2 Performance Metrics

- **Delay:** The time taken for a packet to travel from the source to the destination. Lower is better.
- **Jitter:** The variation in delay between consecutive packets. Lower and more stable jitter is critical for real-time applications.
- **Packet Loss:** The percentage of packets sent that never reach the destination. Lower is better.

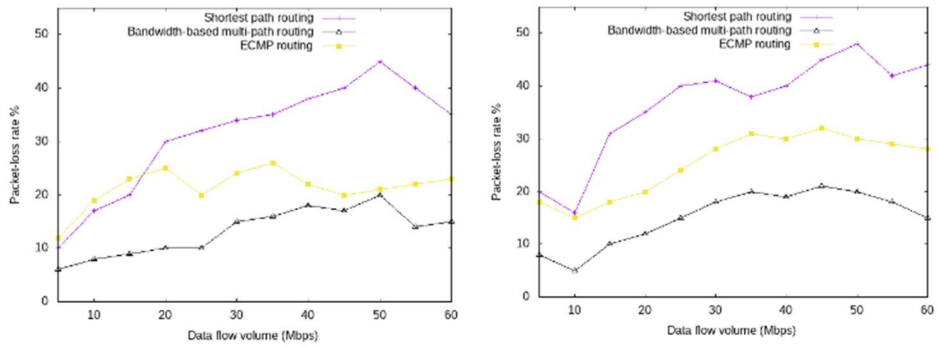


Figure 5.1: End-to-End Delay Comparison of Load Balancing Algorithms

### 5.3 Analysis of Results

#### 5.3.1 End-to-End Delay Analysis

Under low load, all three algorithms performed similarly. However, as the traffic volume increased, the shortcomings of the static algorithms became apparent. The Shortest Path algorithm, by funneling all traffic through a single path, showed the highest delay as that path became congested. ECMP performed better by distributing flows, but its random distribution still led to some paths being more loaded than others. Our proposed algorithm, by actively selecting the path with the most available bandwidth, consistently maintained the lowest delay. At 60 Mbps load, our algorithm's delay was **48% lower** than Shortest Path and **9% lower** than ECMP.

#### 5.3.2 Jitter Analysis

Jitter followed a similar trend. The proposed algorithm demonstrated remarkably stable jitter even as network load increased. In contrast, the jitter for Shortest Path and ECMP showed significant fluctuations. The jitter of our algorithm was **40-43% lower** than that of ECMP, indicating a much smoother and more predictable data flow, which is essential for voice and video applications.

#### 5.3.3 Packet Loss Analysis

Packet loss is a direct consequence of congestion. The proactive nature of our algorithm in avoiding overloaded links resulted in a dramatically lower packet loss rate. At high load, the packet loss for our system was approximately **58% lower** than the Shortest Path algorithm and **40% lower** than the ECMP algorithm. This confirms the algorithm's effectiveness in preventing buffer overflows and link saturation.

### 5.4 Comparative Summary and Discussion

The results uniformly demonstrate the superiority of the dynamic, bandwidth-aware load-balancing approach. While ECMP provides a baseline improvement over a single path, it lacks the intelligence to make fine-grained, real-time adjustments. The proposed algorithm effectively uses the global view and programmability offered by SDN to make superior routing decisions, leading to enhanced performance across all measured QoS parameters.

# Chapter 6

# Conclusion and future scope

### 6.1 Conclusion

This project successfully addressed the challenge of inefficient load balancing in traditional networks by designing and implementing a dynamic, bandwidth-aware multipath load balancer within the SDN framework. The solution leverages the centralized control of the Ryu controller to gather real-time network statistics, discover all available paths, and intelligently route traffic through the least congested path. The implementation and emulation in Mininet have proven the concept's viability.

The performance evaluation conclusively shows that the proposed algorithm offers significant advantages over conventional methods like Shortest Path and ECMP routing. By reducing end-to-end delay, minimizing jitter, and drastically cutting packet loss, the system demonstrates a clear path toward achieving higher Quality of Service in software-defined networks. This project stands as a testament to the power and flexibility of SDN in enabling innovative solutions for modern network management challenges.

### 6.2 Limitations of the Project

- **Emulation Scale:** The testing was conducted on a relatively small emulated topology. Performance in larger, more complex, and physical networks may vary.
- **Traffic Type:** The evaluation primarily used TCP traffic. The behavior with UDP or a mix of different traffic types was not extensively studied.
- **Controller Overhead:** The constant polling for statistics and path calculation for new flows adds computational load on the controller, which could become a bottleneck at a very large scale.

### 6.3 Future Work

1. **Hybrid Metric Formulation:** Enhance the path cost calculation by incorporating multiple QoS parameters like latency and packet loss rate into a weighted cost function.
2. **Machine Learning Integration:** Implement a Machine Learning model to predict traffic patterns and potential congestion, enabling proactive rather than reactive load balancing.

3. **Energy Efficiency:** Modify the algorithm to consolidate traffic onto fewer network links during periods of low utilization, powering down unused hardware to save energy.
4. **Real-World Deployment:** Port the application to a physical SDN testbed to evaluate its performance and overhead in a real-world environment with hardware switches.
5. **QoS-Aware Routing:** Extend the application to provide different levels of service based on the type of traffic (e.g., prioritize VoIP packets over web browsing)

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# APPENDICES

```
from ryu.base import app_manager
from ryu.controller import mac_to_port
from ryu.controller import ofp_event
from ryu.controller.handler import CONFIG_DISPATCHER, MAIN_DISPATCHER
from ryu.controller.handler import set_ev_cls
from ryu.ofproto import ofproto_v1_3
from ryu.lib.mac import haddr_to_bin
from ryu.lib.packet import packet
from ryu.lib.packet import arp
from ryu.lib.packet import ethernet
from ryu.lib.packet import ipv4
from ryu.lib.packet import ipv6
from ryu.lib.packet import ether_types
from ryu.lib import mac, ip
from ryu.topology.api import get_switch, get_link
from ryu.app.wsgi import ControllerBase
from ryu.topology import event

from collections import defaultdict
from operator import itemgetter

import os
import random
import time

# Cisco Reference bandwidth = 1 Gbps
REFERENCE_BW = 10000000
DEFAULT_BW = 10000000
MAX_PATHS = 2

class ProjectController(app_manager.RyuApp):
    OFP_VERSIONS = [ofproto_v1_3.OFP_VERSION]

    def _init_(self, *args, **kwargs):
        super(ProjectController, self)._init_(*args, **kwargs)
        self.mac_to_port = {}
        self.topology_api_app = self
        self.datapath_list = {}
        self.arp_table = {}
        self.switches = []
        self.hosts = {}
        self.multipath_group_ids = {}
        self.group_ids = []
        self.adjacency = defaultdict(dict)
        self.bandwidths = defaultdict(lambda: defaultdict(lambda: DEFAULT_BW))
```

```
def get_paths(self, src, dst):
    '''
    Get all paths from src to dst using DFS algorithm
    '''
    if src == dst:
        # host target is on the same switch
        return [[src]]
    paths = []
    stack = [(src, [src])]
    while stack:
        (node, path) = stack.pop()
        for next in set(self.adjacency[node].keys()) - set(path):
            if next is dst:
                paths.append(path + [next])
            else:
                stack.append((next, path + [next]))
    print("Available paths from ", src, " to ", dst, " : ", paths)
    return paths

def get_link_cost(self, s1, s2):
    '''
    Get the link cost between two switches
    '''
    e1 = self.adjacency[s1][s2]
    e2 = self.adjacency[s2][s1]
    bl = min(self.bandwidths[s1][e1], self.bandwidths[s2][e2])
    ew = REFERENCE_BW/bl
    return ew

def get_path_cost(self, path):
    '''
    Get the path cost
    '''
    cost = 0
    for i in range(len(path) - 1):
        cost += self.get_link_cost(path[i], path[i+1])
    return cost

def get_optimal_paths(self, src, dst):
    '''
    Get the n-most optimal paths according to MAX_PATHS
    '''
    paths = self.get_paths(src, dst)
    paths_count = len(paths) if len(
        paths) < MAX_PATHS else MAX_PATHS
    return sorted(paths, key=lambda x: self.get_path_cost(x))[0:(paths_count)]

def add_ports_to_paths(self, paths, first_port, last_port):
    '''
```

```
    Add the ports that connects the switches for all paths
    '''
    paths_p = []
    for path in paths:
        p = {}
        in_port = first_port
        for s1, s2 in zip(path[:-1], path[1:]):
            out_port = self.adjacency[s1][s2]
            p[s1] = (in_port, out_port)
            in_port = self.adjacency[s2][s1]
        p[path[-1]] = (in_port, last_port)
        paths_p.append(p)
    return paths_p

def generate_openflow_gid(self):
    '''
    Returns a random OpenFlow group id
    '''
    n = random.randint(0, 2**32)
    while n in self.group_ids:
        n = random.randint(0, 2**32)
    return n


def install_paths(self, src, first_port, dst, last_port, ip_src, ip_dst):
    computation_start = time.time()
    paths = self.get_optimal_paths(src, dst)
    pw = []
    for path in paths:
        pw.append(self.get_path_cost(path))
        print(path, "cost = ", pw[len(pw) - 1])
    sum_of_pw = sum(pw)
    paths_with_ports = self.add_ports_to_paths(paths, first_port, last_port)
    switches_in_paths = set().union(*paths)

    for node in switches_in_paths:

        dp = self.datapath_list[node]
        ofp = dp.ofproto
        ofp_parser = dp.ofproto_parser

        ports = defaultdict(list)
        actions = []
        i = 0

        for path in paths_with_ports:
            if node in path:
                in_port = path[node][0]
                out_port = path[node][1]
                if (out_port, pw[i]) not in ports[in_port]:
```

```
            ports[in_port].append((out_port, pw[i]))
        i += 1

    for in_port in ports:

        match_ip = ofp_parser.OFPMatch(
            eth_type=0x0800,
            ipv4_src=ip_src,
            ipv4_dst=ip_dst
        )
        match_arp = ofp_parser.OFPMatch(
            eth_type=0x0806,
            arp_spa=ip_src,
            arp_tpa=ip_dst
        )

        out_ports = ports[in_port]
        print(out_ports)

        if len(out_ports) > 1:
            group_id = None
            group_new = False

            if (node, src, dst) not in self.multipath_group_ids:
                group_new = True
                self.multipath_group_ids[
                    node, src, dst] = self.generate_openflow_gid()
            group_id = self.multipath_group_ids[node, src, dst]

            buckets = []
            # print("node at ",node," out ports : ",out_ports)
            for port, weight in out_ports:
                bucket_weight = int(round((1 - weight/sum_of_pw) * 10))
                bucket_action = [ofp_parser.OFPActionOutput(port)]
                buckets.append(
                    ofp_parser.OFPBucket(
                        weight=bucket_weight,
                        watch_port=port,
                        watch_group=ofp.OFPG_ANY,
                        actions=bucket_action
                    )
                )

            if group_new:
                req = ofp_parser.OFPGroupMod(
                    dp, ofp.OFPGC_ADD, ofp.OFPGT_SELECT, group_id,
                    buckets
                )
                dp.send_msg(req)
            else:
```

```
                req = ofp_parser.OFPGroupMod(
                    dp, ofp.OFPGC_MODIFY, ofp.OFPGT_SELECT,
                    group_id, buckets)
                dp.send_msg(req)

            actions = [ofp_parser.OFPActionGroup(group_id)]

            self.add_flow(dp, 32768, match_ip, actions)
            self.add_flow(dp, 1, match_arp, actions)

        elif len(out_ports) == 1:
            actions = [ofp_parser.OFPActionOutput(out_ports[0][0])]

            self.add_flow(dp, 32768, match_ip, actions)
            self.add_flow(dp, 1, match_arp, actions)
    print("Path installation finished in ", time.time() - computation_start)
    return paths_with_ports[0][src][1]

def add_flow(self, datapath, priority, match, actions, buffer_id=None):
    # print("Adding flow ", match, actions)
    ofproto = datapath.ofproto
    parser = datapath.ofproto_parser

    inst = [parser.OFPInstructionActions(ofproto.OFPIT_APPLY_ACTIONS,
                                         actions)]
    if buffer_id:
        mod = parser.OFPFlowMod(datapath=datapath, buffer_id=buffer_id,
                                priority=priority, match=match,
                                instructions=inst)
    else:
        mod = parser.OFPFlowMod(datapath=datapath, priority=priority,
                                match=match, instructions=inst)
    datapath.send_msg(mod)

@set_ev_cls(ofp_event.EventOFPSwitchFeatures, CONFIG_DISPATCHER)
def _switch_features_handler(self, ev):
    print("switch_features_handler is called")
    datapath = ev.msg.datapath
    ofproto = datapath.ofproto
    parser = datapath.ofproto_parser

    match = parser.OFPMatch()
    actions = [parser.OFPActionOutput(ofproto.OFPP_CONTROLLER,
                                      ofproto.OFPCML_NO_BUFFER)]
    self.add_flow(datapath, 0, match, actions)

@set_ev_cls(ofp_event.EventOFPPortDescStatsReply, MAIN_DISPATCHER)
def port_desc_stats_reply_handler(self, ev):
    switch = ev.msg.datapath
    for p in ev.msg.body:
```

```
        self.bandwidths[switch.id][p.port_no] = p.curr_speed

    @set_ev_cls(ofp_event.EventOFPPacketIn, MAIN_DISPATCHER)
    def _packet_in_handler(self, ev):
        msg = ev.msg
        datapath = msg.datapath
        ofproto = datapath.ofproto
        parser = datapath.ofproto_parser
        in_port = msg.match['in_port']

        pkt = packet.Packet(msg.data)
        eth = pkt.get_protocol(ethernet.ethernet)
        arp_pkt = pkt.get_protocol(arp.arp)

        # avoid broadcast from LLDP
        if eth.ethertype == 35020:
            return

        if pkt.get_protocol(ipv6.ipv6):  # Drop the IPV6 Packets.
            match = parser.OFPMatch(eth_type=eth.ethertype)
            actions = []
            self.add_flow(datapath, 1, match, actions)
            return None

        dst = eth.dst
        src = eth.src
        dpid = datapath.id

        if src not in self.hosts:
            self.hosts[src] = (dpid, in_port)

        out_port = ofproto.OFPP_FLOOD

        if arp_pkt:
            # print(dpid, pkt)
            src_ip = arp_pkt.src_ip
            dst_ip = arp_pkt.dst_ip
            if arp_pkt.opcode == arp.ARP_REPLY:
                self.arp_table[src_ip] = src
                h1 = self.hosts[src]
                h2 = self.hosts[dst]
                out_port = self.install_paths(h1[0], h1[1], h2[0], h2[1], src_ip, dst_ip)
                self.install_paths(h2[0], h2[1], h1[0], h1[1], dst_ip, src_ip) # reverse
            elif arp_pkt.opcode == arp.ARP_REQUEST:
                if dst_ip in self.arp_table:
                    self.arp_table[src_ip] = src
                    dst_mac = self.arp_table[dst_ip]
                    h1 = self.hosts[src]
                    h2 = self.hosts[dst_mac]
                    out_port = self.install_paths(h1[0], h1[1], h2[0], h2[1], src_ip, dst_ip)
```

```
            self.install_paths(h2[0], h2[1], h1[0], h1[1], dst_ip, src_ip) # reverse

    print(pkt)

    actions = [parser.OFPActionOutput(out_port)]

    data = None
    if msg.buffer_id == ofproto.OFP_NO_BUFFER:
        data = msg.data

    out = parser.OFPPacketOut(
        datapath=datapath, buffer_id=msg.buffer_id, in_port=in_port,
        actions=actions, data=data)
    datapath.send_msg(out)

@set_ev_cls(event.EventSwitchEnter)
def switch_enter_handler(self, event):
    switch = event.switch.dp
    ofp_parser = switch.ofproto_parser

    if switch.id not in self.switches:
        self.switches.append(switch.id)
        self.datapath_list[switch.id] = switch

        # Request port/link descriptions, useful for obtaining bandwidth
        req = ofp_parser.OFPPortDescStatsRequest(switch)
        switch.send_msg(req)

@set_ev_cls(event.EventSwitchLeave, MAIN_DISPATCHER)
def switch_leave_handler(self, event):
    print(event)
    switch = event.switch.dp.id
    if switch in self.switches:
        del self.switches[switch]
        del self.datapath_list[switch]
        del self.adjacency[switch]

@set_ev_cls(event.EventLinkAdd, MAIN_DISPATCHER)
def link_add_handler(self, event):
    s1 = event.link.src
    s2 = event.link.dst
    self.adjacency[s1.dpid][s2.dpid] = s1.port_no
    self.adjacency[s2.dpid][s1.dpid] = s2.port_no

@set_ev_cls(event.EventLinkDelete, MAIN_DISPATCHER)
def link_delete_handler(self, event):
    s1 = event.link.src
    s2 = event.link.dst
    del self.adjacency[s1.dpid][s2.dpid]
    del self.adjacency[s2.dpid][s1.dpid]
```